

v1.0Pion Deep Dive_ One-Pion Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Unified Pion Field Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Pion Deep Dive_ One-Pion Universe V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17} + particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Form factor precision	General VMD model	JLab CLAS12 2024: F_π measured 3-5% precision, $Q^2=0.6-2.5 \text{ GeV}^2$. Smooth VMD/pQCD interpolation added	MEDIUM	Improved form factor changes pion electromagnetic coupling strength
F_π chiral condensate	Approximate value	FLAG 2024: $F_\pi = 92.07 \pm 0.46 \text{ MeV}$ (0.5% precision). V2.0: $\pi \rightarrow \gamma \gamma$ decay rate precisely updated	LOW	Sub-percent pion decay constant improves chiral EFT predictions
Pionium bound state	Not modeled	DIRAC: pionium $\tau = 2.91 \pm 0.17 \times 10^{-15} \text{ s}$. V2.0: PIONIUM bound state of $\pi^+\pi^-$ worldlines added	MEDIUM	New composite system — $\pi^+\pi^-$ atom — absent from V1.0

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Quark mass dependence	Delta_m_pi = F(m_q) noted	FLAG 2024: m_u=2.16, m_d=4.67 MeV. Pion mass computed dynamically from quark masses	LOW	Dynamic mass calculation for changed quark mass parameters

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** See table
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Pion Deep Dive_ One-Pion Universe V1.0 archived | V2.0 is the active specification as of June 2026